

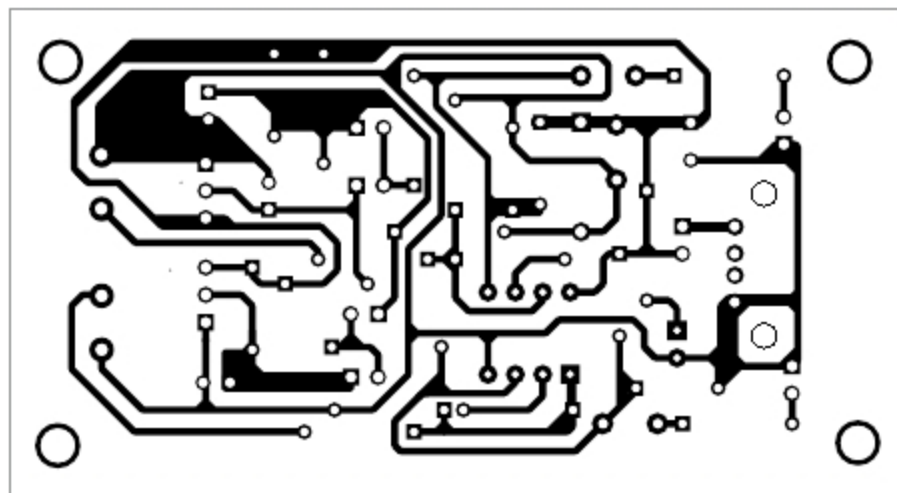


Fig. 2: Actual-size PCB layout for dual level shifter and buffer

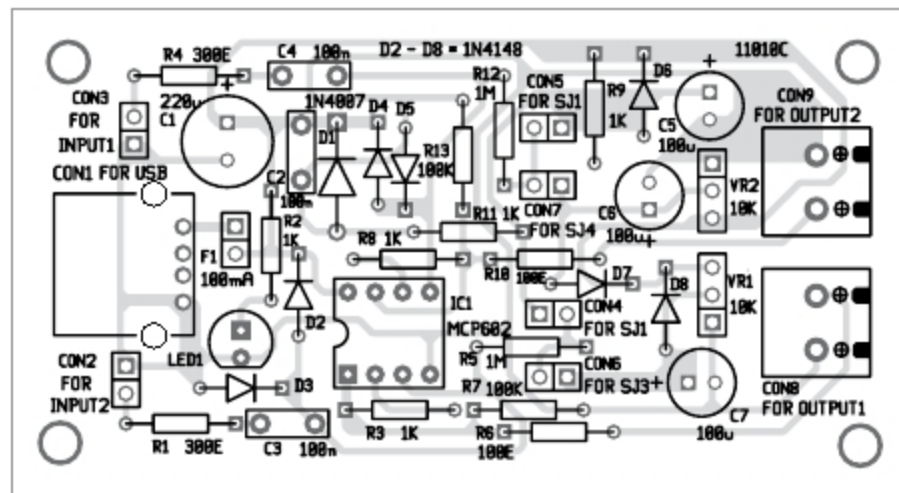


Fig. 3: Components layout for PCB

Capacitors C3 and C4 cut the input DC components (if any). Resistors R1 and R4 and diodes D2 through D5 protect the inputs of operational amplifier IC1. Resistors R5 and R12 are used to apply the DC offset to the input of op-amps A1 and A2. For each op-amp, there are two independent options for the DC offset. First is fixed offset, which is equal to $V_{cc}/2$. Second is adjustable offset.

The table gives the selection of the DC offsets for the level shifters with shorting jumpers.

Shorting jumpers SJ1 and SJ2

should not be closed at the same time. Similarly, SJ3 and SJ4 should also not be closed at the same time. The circuit is operational in full power supply range of the op-amps. Use rail-to-rail input and output op-amps.

Construction and testing

An actual-size PCB layout for the dual level shifter and buffer is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, connect the USB connector for 5V (V_{cc}) regulated power supply. Connect potentiometers VR1 and

VR2 on the front panel of the cabinet for variable offsets. Connect LED1 on the front panel to indicate power status. Use shorting jumpers according to requirement, as shown in the table. **EFY**



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EYSGCNZXY EYSGJNZXY EYAGJNZXX



BT4.2, STD, Antenna, 32KHz clock DC/DC
BT4.2, Small, Antenna
BT4.2+ANT™, Small, Antenna

BT5.0 - Cortex-M4/M4F (nRF52)

EYSHCNZWX EYSHJNZWX EYSHSNZWX



BT5.0+ANT™, STD, Antenna, 32KHz clock DC/DC
BT5.0+ANT™, Small, Antenna
BT5.0+ANT™, Ultra Small, Antenna

EYSKBNZWB EYSLCNZWW



Full BT5.0+802.15.4 (Thread) High performance
BT5.0, STD, Antenna, Simple

Under Development

Local Area Network

WLAN: IEEE802.11 SDIO

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WYSAGVDX7 11b/g/n, Low-P, Antenna

2.4GHz+5GHz (11a/n/ac)

WYS8HVXGX 1x1 11ac, BT4.2, SMD

WYSAGVDXG WYSEGVDXG



1x1 11ac, BT4.2, SMD w/ Antenna
Under Development
1x1 11ac, BT4.2, SMD w/ RF connector

WLAN: IEEE802.11 SW Embedded

WYSACVLXY-XX



11b/g/n + CPU, SMD, Antenna, Cortex-M4, TY app.

Navigation System

GPS/GNSS

GYSFFMAXB



GPS with CPU, Memory, SMD

GYSFFMAXC



GNSS with CPU, Memory, SMD



GYSFDMAXB

GPS with CPU, Memory, Antenna,

- Production
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